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Multivalued Attributes:-

i) make a table of the attribute

Dept-locations	
Dno	Location
1	Paris
1	Sargodha
2	London
2	Lahore

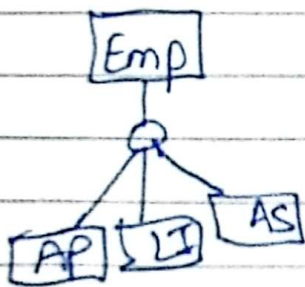
Mapping of N-ary Relationship:- $N \geq 2$

- Create a new table
- PK composite key of all 3 tables.

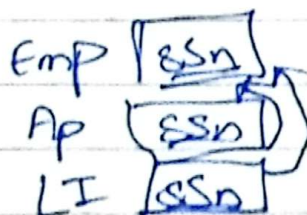
Enhanced ERD:

Mapping Specialization/Generalization:-

a) Multiple relations - Superclass & subclass

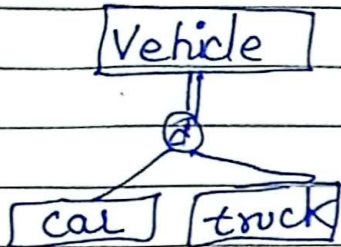


- * Specialization/Gen
- * disjoint/overlapping
- * total/partial



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b) Multiple Relations - Subclass (necessary to have total participation)



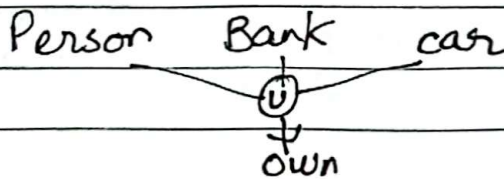
car [VID]
truck [VID]

} Separately cater relationship of vehicle.

c) Single relation - one type attribute
(disjoint case) (discriminating attribute)
(Too many nulls)

d) Single relation - multiple attributes
^{more preferred}
(overlapping + disjoint)
(keep flags) (Too many nulls)

Union :



Person
Id own Id
Bank
Id own-Id
Car
Id own-Id

Surrogate key
Own
own-Id



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Alias (nick name)

Distinct (no duplicates) if two columns then check unique pairs.

Set Operations (multiple tables)
Union, Intersection

is Not } NULL COUNT(*) ⇒ caters NULL
is }
where } COUNT(age) ⇒ doesn't caters

Group by used by aggregate functions.
Having

Condition on aggregate is done by "Having".
orderby

By default order by is "asc"
if two columns then check
for same then go for the other

where
group by
having
order by

In sub-queries, the inner query runs independently

Joins :

Cross Join

Inner join

Natural Join

Outer join

↳ left, right, full outer



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Cross Join: (cross product) (possible combination)

eid	name	salary
1	A	23
2	B	24

eid	dname	relation
2	S	d
2	D	s

eid	name	sal	eid	dname	relation
1	A	23	2	S	d
2	B	24	2	S	d
1	A	23	2	D	s
2	B	24	2	D	s

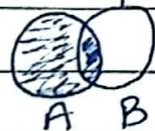
Inner Join: (filter common attribute data) (specifying common attribute)
(common shown twice)

eid	name	sal	eid	dname	rel
2	A	24	2	B	d
2	B	24	2	D	s

Natural Join: (used in relational algebra) (does not specify common attribute)
(limited support)
(only show one)

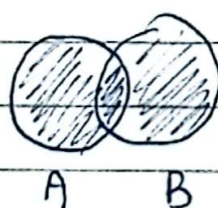
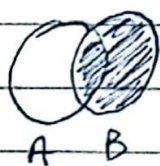
Outer Join: (attribute repeat)

↳ Left OJ:



↳ Full OJ

↳ Right OJ:



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Relational Data Model:-

* Cardinality: Num of rows

* Degree: Num of attributes

STUDENT (ID, name, NIC)
Table (attributes)

t_2 [DOB] = t_2 .DOB
 $t_{\text{row no}}$ [attribute name]

* Relation: Subset of cartesian product of all domains (ordered pairs)

$$R(\gamma) \subseteq \text{Dom}(A_1) \times \text{Dom}(A_2)$$

$$A_1 = (1, 2)$$

$$A_2 = (x, y)$$

$$\gamma = (1, x) (1, y) (2, x) (2, y)$$

$$\text{Dom (Attribute)} = \{DS, CS\}$$

* Keys: Uniquely identify a row.

i) Superkey

iv) Alternate key

ii) Candidate key

v) Unique key

iii) Primary key

vi) Foreign key

• Superkey: $\rightarrow$ Superset $\rightarrow$ multiple attributes $\rightarrow$ repeating attributes
 $\rightarrow$ unique $\rightarrow$ Nulls allow $\rightarrow$ multiple keys
 $\rightarrow$ {name}, {name, ID}

• Candidate key

$\rightarrow$ Minimal

$\rightarrow$ Unique

$\rightarrow$ Nulls allowed

$\rightarrow$ No attribute repeat.

every candidate key is a super key.



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- Primary key

→ only one → unique → null not allowed

- Alternate key

→ left out of primary keys.

- Unique key

→ unique nulls allowed

- Foreign key

→ Primary key of another table

- Surrogate key

→ Artificial column introduced

→ Primary key

→ meaningless to user

Integrity constraints:-

i) Entity

iii) Domain

ii) Referential

iv) Additional

v) Table

Entity

Referential

→ PK (unique, not null)

→ FK that does not exist as a original PK in original table.

→ delete X (cascade child ↔ parent delete ref. from)
→ Update X (No action Null Default)

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Domain constraint
→ CHECK (age < 16)

Attribute constraint
→ same data type.

Normalization

o Informal guidelines

Guideline i) Imparting clear semantics to attributes in relations! Do not mix attributes of 2 entity! Foreign key can exist.

problems:

- a) Info repeat (entity integrity) (consistency issue)
- b) anomalies (Insert, Delete, Update)
(one delete cause significant loss) other table data delete

Guideline ii) Make sure no anomaly of guideline 1

Guideline iii) Few NULL values (As much as possible)

"Null" could have multiple meanings.

→ attribute not applicable to row!

Name	HOP
Hamad	Null
Huma	Hamad

→ Value of attribute is unknown (never tell in future)

SSN	blood
-	Null

→ unknown → Absent but will be known in future.

SSN	DNo
9237	-



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Functional dependency :-

$X \rightarrow Y$ (X determines Y)
(determine) (depend)

e.g., Roll number determines Name of student

- i) Reflexive $X \rightarrow Y$ (subset of X)
- ii) Argumentation $XZ \rightarrow YZ$
- iii) Transitive

→ identifies itself
→ adding an extra attribute

Decomposition: $X \rightarrow YZ$; $X \rightarrow Y$ and $X \rightarrow Z$

Union : $X \rightarrow Y$, $X \rightarrow Z$; $X \rightarrow YZ$

Pseudotransitivity : $X \rightarrow Y$
 $WY \rightarrow Z$
 $WX \rightarrow Z$

Transitive : $X \rightarrow Y \rightarrow Z$
 $Y \rightarrow Z$

Example: 1

			2	
	$A \rightarrow B$	$D \rightarrow E$		$A \rightarrow E$
$A \rightarrow BC$	$B \rightarrow CD$	$A \rightarrow CD$		
$A \rightarrow B$	$A \rightarrow C$	$CD \rightarrow EF$	$A \rightarrow D$	
$B \rightarrow E$	$AD \rightarrow CD$		$D \rightarrow E$	
$A \rightarrow E$	$A \rightarrow E$		$A \rightarrow E$	
	$AD \rightarrow EF$			



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$$X \rightarrow YZ$$

$$X \rightarrow Y$$

$$Y \rightarrow W$$

$$Y \rightarrow W$$

$$ZW \rightarrow V$$

$$X \rightarrow W$$

$$XZ \rightarrow WZ$$

~~$$ZX \rightarrow V$$~~

$$ZW \rightarrow V$$

$$XZ \rightarrow V$$

Types of FDs:

with Composite determinant

* Full functional dependency

* Partial functional dependency

e.g., $\underbrace{XZ}_{\text{composite}} \rightarrow Y$

→ $XZ \rightarrow Y$ (Y is fully dependent on composite det)

→ $XZ \rightarrow Y$ (any one can also be enough)